

The Stable-Rank Matrix Extraction in arXiv:1605.03861

An algorithm was already supplied by arXiv:1507.02268

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Status. Literature already answered. Cohen–Nelson–Woodruff give a deterministic polynomial-time algorithm with exactly the stable-rank support size and additive spectral error requested in the diagonal formulation of the source.

1 Source question

Friedland and Youssef prove that for every $A \in \mathbb{R}^{n \times m}$ and $\varepsilon > 0$ there is a nonnegative diagonal matrix D with at most $O(\text{srnk}(A)/\varepsilon^2)$ nonzero entries such that

$$\|ADA^* - AA^*\| \leq \varepsilon \|A\|^2, \quad \text{srnk}(A) = \frac{\|A\|_F^2}{\|A\|^2}. \quad (1)$$

Immediately after their Theorem 1.1 they explicitly identify this as an equivalent diagonal formulation of their reweighted column extraction. In the introduction they note that their proof is existential and say they do not know how to produce an algorithm attaining it [1].

2 Answering theorem

Cohen, Nelson, and Woodruff prove the following result in their Section 4 [2, Theorem 5].

Theorem 1 (Cohen–Nelson–Woodruff). *Let $B \in \mathbb{R}^{N \times d}$ satisfy $\|B\|^2 \leq 1$ and $\|B\|_F^2 \leq k$, and let $0 < \varepsilon < 1$. There exists a diagonal matrix S with $O(k/\varepsilon^2)$ nonzero entries such that*

$$\|(SB)^T(SB) - B^T B\| \leq \varepsilon. \quad (2)$$

Moreover, S can be computed by a deterministic polynomial-time algorithm.

Their proof is a stable-rank extension of the Batson–Spielman–Srivastava barrier method. It is constructive: the final paragraph verifies that all barrier invariants and step sizes can be evaluated in polynomial time.

3 One-line reduction

Assume $A \neq 0$ and apply the answering theorem to

$$B = \frac{A^*}{\|A\|} \in \mathbb{R}^{m \times n}, \quad k = \lceil \text{srnk}(A) \rceil.$$

Then $\|B\| = 1$ and $\|B\|_F^2 = \text{srnk}(A) \leq k$. Put $D = S^2$. Since S is diagonal, D is nonnegative diagonal with the same support, and (2) becomes

$$\left\| \frac{AS^2A^*}{\|A\|^2} - \frac{AA^*}{\|A\|^2} \right\| \leq \varepsilon.$$

Multiplication by $\|A\|^2$ is exactly (1). A nonzero matrix has stable rank at least one, so replacing it by its ceiling changes only the absolute constant in the support bound. The zero matrix is trivial.

Thus the source's diagonal extraction can be produced deterministically in polynomial time with the same optimal-order $O(\text{srnk}(A)/\varepsilon^2)$ support.

4 Chronology and scope

The chronology is unusual. Version 3 of the answering preprint was posted on 2 March 2016. Version 1 of the source was posted on 12 May 2016. The answer therefore predates the question by about ten weeks; its ICALP publication appeared in August 2016.

The reduction resolves exactly the arbitrary nonnegative diagonal-weight formulation that Friedland–Youssef state as equivalent to their extraction conclusion. Cohen–Nelson–Woodruff do not require the selected weights to be equal. Accordingly, this packet does not claim a stronger algorithm in which every weight is equal or an integer multiple of a common quantum.

References

- [1] O. Friedland and P. Youssef, *Approximating matrices and convex bodies through Kadison–Singer*, arXiv:1605.03861 (2016).
- [2] M. B. Cohen, J. Nelson, and D. P. Woodruff, *Optimal approximate matrix product in terms of stable rank*, 43rd ICALP, LIPIcs 55 (2016), Article 11; arXiv:1507.02268v3.